Chapter 4, Section 4

James Lee

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3. Let the elliptic curve X be embedded in $\mathbf{P}^2$ so as to have the equation $y^2 = x(x-1)(x-\lambda)$. Show that any automorphism of X leaving $P_0 = (0:1:0)$ fixed is induced by an automorphism of $\mathbf{P}^2$ coming from the automorphism of the affine (x, y)-plane given by

$$\begin{cases} x' = ax + b \\ y' = cy. \end{cases}$$

In each of the four cases of (4.7), describe these automorphisms of $\mathbf{P}^2$ explicitly, and hence determine the structure of the group $G = \operatorname{Aut}(X, P_0)$.

Proof. The embedding of X in $\mathbf{P}^2$ is given by the complete linear system $|3P_0| \cong \mathbf{P}^2$. Let $1, x, y$ be a k-basis of $\mathrm{H}^0(X, \mathcal{O}_X(3P_0))$ such that x, y are local affine coordinates of $|3P_0|$. Any automorphism of X leaving P_0 induces an automorphism of $|3P_0|$, i.e., there exists a faithful representation $\rho : G \hookrightarrow \operatorname{Aut}(|3P_0|) \cong \mathrm{PGL}(2)$. If $\sigma \in G$, then we can write

$$\rho(\sigma) = \begin{pmatrix} a & 0 & b \\ c & d & e \\ f & 0 & g \end{pmatrix}, \quad d(ag - bf) \neq 0.$$

The unique g^1_2, denoted $f : X \to \mathbf{P}^1$, is given by the projection of X onto the x-axis. By (4.4), there exists $\tau \in \mathrm{PGL}(1)$ such that $\tau \circ f = f \circ \sigma$.

$\boxed{j \neq 0, 1728}$ The unique nontrivial element $\sigma \in G$ interchanges the sheets of the unique g^1_2 and can be written as

$$\begin{cases} x' = x \\ y' = -y \end{cases}.$$

$\boxed{j = 1728 \text{ and } \operatorname{char} k \neq 3}$

$\boxed{j = 0 \text{ and } \operatorname{char} k \neq 3}$

$\boxed{j = 0\ (= 1728) \text{ and } \operatorname{char} k = 3}$ □

4. Let X be an elliptic curve in $\mathbf{P}^2$ given by an equation of the form

$$y^2 + a_1 xy + a_3 y = x^3 + a_2 x^2 + a_4 x + a_6.$$

Show that the j-invariant is a rational function of the a_i, with coefficients in $\mathbf{Q}$. In particular, if the a_i are all in some field $k_0 \subseteq k$, then $j \in k_0$ also. Furthermore, for every $\alpha \in k_0$, there exists an elliptic curve defined over k_0 with j-invariant equal to α.

Proof. Complete the square in y to get an equation of the form

$$y^2 = x^3 + Ax^2 + Bx + C,$$

with

$$A = \frac{a_1^2}{4} + a_2, \quad B = \frac{a_1 a_3}{2} + a_4, \quad C = \frac{a_3^2}{4} + a_6.$$

Let α, β, γ be the roots of the cubic on the right-hand side. Make a linear change of x to send α, β to $0, 1$ so that the equation becomes

$$y^2 = x(x-1)(x-\lambda), \quad \lambda = \frac{\gamma - \alpha}{\beta - \alpha}.$$

Plugging in λ into the j-invariant formula gives

$$j = -2^8 \frac{(\alpha+\beta+\gamma)^2 - 3(\alpha\beta - \beta\gamma - \gamma\alpha)}{(\alpha-\beta)^2(\beta-\gamma)^2(\gamma-\alpha)^2}.$$

This is a rational function of the symmetric polynomials in α, β, γ, which are themselves polynomials in A, B, C, and hence in the a_i. $\square$

7. *The Dual of a Morphism.* Let X and X' be elliptic curves over k, with base points P_0, P_0'.

(a) If $f : X \to X'$ is any morphism, use (4.11) to show that $f^* : \operatorname{Pic} X' \to \operatorname{Pic} X$ induces a homomorphism $\hat{f} : (X', P_0') \to (X, P_0)$. We call this the *dual* of f.

(b) If $f : X \to X'$ and $g : X' \to X''$ are two morphisms, then $\widehat{g \circ f} = \hat{f} \circ \hat{g}$.

(c) Assume $f(P_0) = P_0'$, and let $n = \deg f$. Show that if $Q \in X$ is any point, and $f(Q) = Q'$, then $\hat{f}(Q') = n_X(Q)$. Conclude that $f \circ \hat{f} = n_{X'}$ and $\hat{f} \circ f = n_X$.

(d) If $f, g : X \to X'$ are two morphisms preserving the base points P_0, P_0', then $\widehat{f+g} = \hat{f} + \hat{g}$.

(e) Using (d), show that for any $n \in \mathbf{Z}$, $\hat{n}_X = n_X$. Conclude that $\deg n_X = n^2$.

(f) Show for any f that $\deg \hat{f} = \deg f$.

Proof.

(a) Any morphism $f : X \to X'$ induces a natural transformation of functors $f^* : \operatorname{Pic}^\circ(X'/-) \to \operatorname{Pic}^\circ(X/-)$ (4.10.2). Under the identification $\operatorname{Pic}^\circ(X/-) \cong \operatorname{Hom}(-, J)$ and $\operatorname{Pic}^\circ(X'/-) \cong \operatorname{Hom}(-, J')$, Yoneda's lemma says there exists a natural bijection

$$\operatorname{Hom}(J', J) \cong \operatorname{Hom}(\operatorname{Pic}^\circ(X'/-), \operatorname{Pic}^\circ(X/-)),$$

which means there exists a unique $\hat{f} : J' \to J$ such that f^* is given by composition by $\hat{f}$. We obtain our desired morphism by identifying $X \cong J$ and $X' \cong J'$, and composing $\hat{f}$ by an automorphism of X that sends $f(P_0')$ to P_0 (4.2).

(b)

(c)

(d)

(e) Apply (d) to $f = \mathrm{id}_X$ and $\hat{1}_X = 1_X$ to get $\hat{n}_X = n_X$.

(f)

$\square$

9. We say two elliptic curves X, X' are *isogenous* if there is a finite morphism $f : X \to X'$.

(a) Show that isogeny is an equivalence relation.

(b) For any elliptic curve X, show that the set of elliptic curves X' isogenous to X, up to isomorphism is countable.

Proof.

(b)

$\square$

11. Let X be an elliptic curve over $\mathbf{C}$, defined by the elliptic functions with periods $1, \tau$. Let R be the the ring of endormorphisms of X.

(a) If $f \in R$ is a nonzero endormorphism correspondending to complex multiplication by α as in (4.18), show that $\deg f = |\alpha|^2$.

(b) If $f \in R$ corresponds to $\alpha \in \mathbf{C}$ again, show that the dual $\hat{f}$ of (Ex. 4.7) corresponds to the complex conjugate $\bar{\alpha}$ of α.

(c) If $\tau \in \mathbf{Q}(\sqrt{-d})$ happens to be integral over $\mathbf{Z}$, show that $R = \mathbf{Z}[\tau]$.

Proof.

(a)

(b)

(c)

□

13. If $p = 13$, there is just one value of j for which the Hasse invariant of the corresponding curve is 0. Find it.

Proof. The polynomial $h_p(\lambda)$ of (4.22) for $p = 13$ is

$$h_{13}(\lambda) = \lambda^6 + 10\lambda^5 + 4\lambda^4 + 10\lambda^3 + 4\lambda^2 + 10\lambda + 1,$$

and an elliptic curve has Hasse invariant 0 if and only if $h_{13}(\lambda) = 0$. On the otherhand, consider the j-invariant

$$j = 2^8 \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}.$$

Expanding above modulo 13 gives

$$\lambda^6 + 10\lambda^5 + (10j + 6)\lambda^4 + (6j + 6)\lambda^3 + (10j + 6)\lambda^2 + 10\lambda + 1 = 0.$$

Matching coefficients with $h_{13}(\lambda)$ and solving for j gives $j \equiv 5 \mod 13$. □

14. The Fermat curve $X : x^3 + y^3 = z^3$ gives a nonsingular curve in characteristic p for every $p \neq 3$. Determine the set $\mathfrak{P} = \{p \neq 3 \mid X_{(p)} \text{ has Hasse invariant } 0\}$, and observe (modulo Dirichlet's theorem) that it is a set of primes of density $1/2$.

Proof. Using (4.21) with $f(x, y, z) = x^3 + y^3 - z^3$, we find that the coefficient of $(xyz)^{p-1}$ in f^{p-1} is zero if and only if $p \equiv 2 \mod 3$. By Dirichlet's theorem, the density of $\mathfrak{P}$ is $1/\phi(3) = 1/2$. □

15. Let X be an elliptic curve over a field k of characteristic p. Let $F' : X_p \to X$ be the k-linear Frobenius morphism (2.4.1). Use (4.10.7) to show that the dual morphism $\hat{F}' : X \to X_p$ is separable if and only if the Hasse invariant of X is 1. Now use (Ex. 4.7) to show that if the Hasse invariant is 1, then the subgroup of points of order p on X is isomorphic to $\mathbf{Z}/p$; if the Hasse invariant is 0, then it is 0.

Proof. □

16. Again let X be an elliptic curve over k of characteristic p, and suppose X is defined over the field $\mathbf{F}_q$ of $q = p^r$ elements, i.e., $X \subseteq \mathbf{P}^2$ can be defined by an equation with coefficients in $\mathbf{F}_q$. Assume also that X has a rational point over $\mathbf{F}_q$. Let $F' : X_q \to X$ be the k-linear Frobenius with respect to q.

(a) Show that $X_q \cong X$ as schemes over k, and that under this identification, $F' : X \to X$ is the map obtained by the qth-power map on the coordinates of points of X, embedded in $\mathbf{P}^2$.

(b) Show that $1_X - F'$ is a separable morphism and its kernel is just the set $X(\mathbf{F}_q)$ of points of X with coordinates in $\mathbf{F}_q$.

(c) Using (Ex. 4.7), show that $F' + \hat{F}' = a_X$ for some integer a, and that $N = q - a + 1$, where $N = \#X(\mathbf{F}_q)$.

(d) Use the fact that $\deg(m + nF') > 0$ for all $m, n \in \mathbf{Z}$ to show that $|\alpha| \leq 2\sqrt{q}$. This is Hasse's proof of the analogue of the Riemann hypothesis for elliptic curves.

(e) Now assume $q = p$, and show that the Hasse invariant of X is 0 if and only if $a \equiv 0 \mod p$. Conclude that for $p \geq 5$ that X has Hasse invariant 0 if and only if $N = p + 1$.

Proof.

(a)

(b)

(c)

(d)

(e)

□

17. Let X be the curve $y^2 + y = x^3 - x$ of (4.23.8).

(a) If $Q = (a, b)$ is a point on the curve, compute the coordinates of the point $P + Q$, where $P = (0, 0)$, as a function of a, b. Use this formula to find the coordinates nP, $n = 1, 2, \ldots, 10$.

(b) This equation defines a nonsingular curve over $\mathbf{F}_q$ for all $p \neq 37$.

Proof.

(a)

(b)

□

20. Let X be an elliptic curve over a field k of characteristic $p > 0$, and let $R = \mathrm{End}(X, P_0)$ be its ring of endormorphisms.

(a) Let X_p be the curve over k defined by changing the k-structure of X (2.4.1). Show that $j(X_p) = j(X)^{1/p}$. Thus, $X \cong X_p$ over k if and only if $j \in \mathbf{F}_p$.

(b) Show that p_X in R factors into a product of $\pi\hat{\pi}$ of two elements of degree p if and only if $X \cong X_p$. In this case, the Hasse invariant of X is 0 if and only if π and $\hat{\pi}$ are associates in R.

(c) If $\mathrm{Hasse}(X) = 0$ show in any case $j \in \mathbf{F}_{p^2}$.

(d) For any $f \in R$, there is an induced map $f^* : \mathrm{H}^1(\mathcal{O}_X) \to \mathrm{H}^1(\mathcal{O}_X)$. This must be multiplication by an element $\lambda_f \in k$. So we obtain a ring homomorphism $\varphi : R \to k$ by sending f to λ_f. Show that any $f \in R$ commutes with the (nonlinear) Frobenius moprhism $F : X \to X$, and conclude that if $\mathrm{Hasse}(X) \neq 0$, then the image of φ is $\mathbf{F}_p$. Therefore, R contains a prime ideal $\mathfrak{p}$ with $R/\mathfrak{p} \cong \mathbf{F}_p$.

Proof.

(a)

(b)

(c)

(d)

□

22. If $X \to \mathbf{A}^1_C$ is a family of elliptic curves having a section, show that the family is trivial.

Proof. □